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FORM TP 2007048

MAY/JUNE 2007

CARIBBEAN EXAMINATIONS COUNCIL

SECONDARY EDUCATION CERTIFICATE EXAMINATION

BIOLOGY

Paper 02 – General Proficiency

$1\frac{1}{2}$ hours

READ THE FOLLOWING DIRECTIONS CAREFULLY

1. **There are FIVE questions in this booklet. Answer ALL questions.**
2. **You MUST use this answer booklet when responding to the questions. For each question, write your answer in the space provided and return the answer booklet at the end of the examination.**

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NOTHING HAS BEEN OMITTED

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Answer ALL questions.

You should NOT spend more than 30 minutes on Question 1.

1. Students in a Biology class wish to test the hypothesis that all leaves on a tree photosynthesize at the same rate. They select leaves from a tree similar to that illustrated in Figure 1.

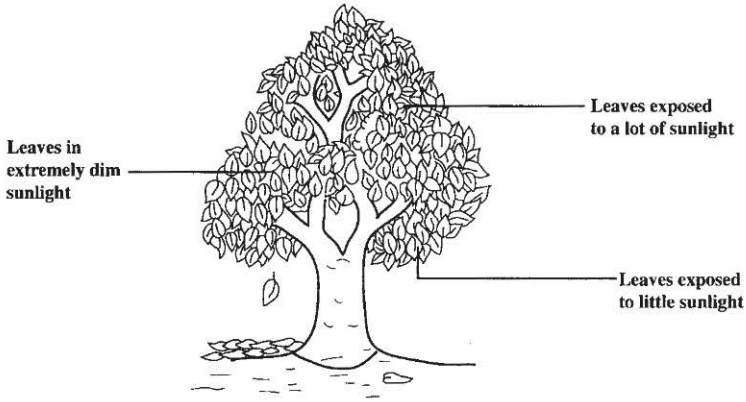

Figure 1. Diagram of a tree displaying its canopy

(a) (i) Describe how the students may investigate the hypothesis. Include in your description any apparatus/materials required.

Apparatus/materials:

_____

_____

Method:

_____

_____

_____

_____

_____

(6 marks)

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(ii) Write an aim for the investigation.

_____

_____

(2 marks)

(iii) Suggest ONE precaution the students should take in their investigation.

_____

(1 mark)

(b) Table 1 below shows the effect of carbon dioxide concentration on the rate of photosynthesis at two different light intensities.

TABLE 1: EFFECT OF CARBON DIOXIDE ON PHOTOSYNTHESIS

CO_2 concentration (%)	Rate of photosynthesis	
	Low light intensity	High light intensity
0.00	0	0
0.02	15	30
0.04	24	50
0.06	31	64
0.08	36	70
0.10	38	74
0.12	40	75
0.14	40	75
0.20	40	75

(i) On the graph paper on page 5, construct a line graph to show the effect of carbon dioxide concentration on the rate of photosynthesis. **(6 marks)**

(ii) Using Table 1 and the graph you constructed, determine the concentration at which carbon dioxide is a limiting factor at high and low light intensities respectively.

At low intensity:

_____

At high intensity:

_____

(2 marks)

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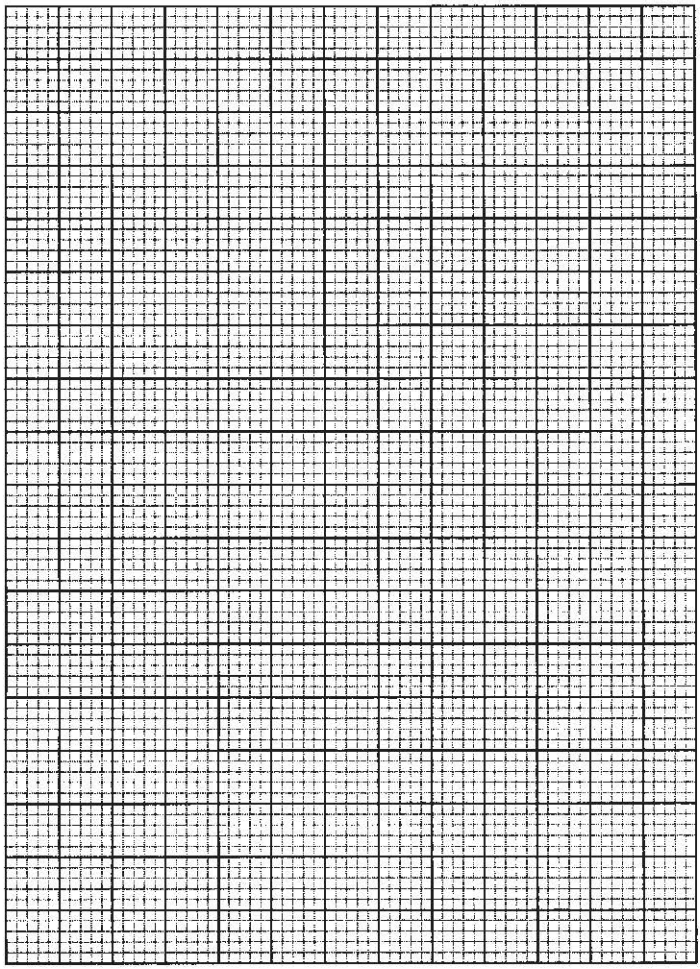

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(iii) Apart from carbon dioxide and sunlight, identify ONE OTHER environmental factor that is required for photosynthesis.

_____

(1 mark)

(c) (i) Using arrows only, illustrate on Figure 2 the pathway by which the products of photosynthesis arrive at storage cells or organs. **(2 marks)**

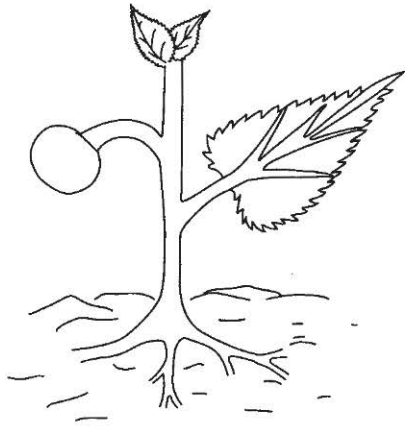

Figure 2. Stylised diagram of the components of a plant

(ii) What is the name of the transporting tissue?

_____

(1 mark)

(d) Table 2 below shows the composition of human breast milk, UHT cow's milk and infant formula.

TABLE 2: COMPOSITION OF THREE TYPES OF MILK

Constituent	Amount / 100g milk		
	Breast milk	**UHT cow's milk**	**Infant formula**
Energy (kcal)	77.0	65.0	100
Water (g)	85.2	87.4	0.0
Protein (g)	1.1	3.5	1.5
Fat (g)	4.0	3.5	4.3
Lactose (milk sugar) (g)	9.5	4.9	9.2
Vitamin C (mg)	5.0	1.0	5.0
Calcium (mg)	33.0	118.0	7.0

(i) Breast milk is described as best for babies. Using the information in Table 2, give TWO reasons why this description is appropriate.

_____

_____

_____

_____

(4 marks)

(ii) Apart from nutrients, suggest ONE OTHER value of breast milk.

_____

(1 mark)

(e) Minerals are required for good nutrition in plants. Name TWO minerals used by plants and state the function of EACH mineral named.

Mineral 1:

_____

Function:

_____

Mineral 2:

_____

Function:

_____

(4 marks)

Total 30 marks

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2. (a) Figure 3 is a diagram of the elbow joint.

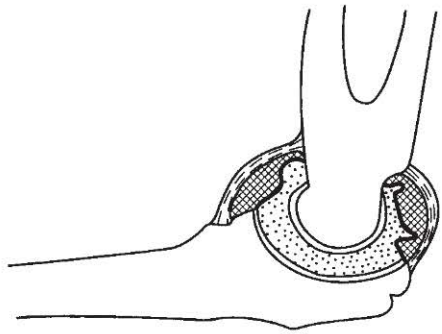

Figure 3. Diagram showing the elbow joint

(i) Name the type of joint illustrated in Figure 3.

_____

(1 mark)

(ii) Label the diagram to illustrate the MAIN features of a joint. **(4 marks)**

(iii) In the diagram at Figure 3, a pad of fat is shown. Suggest ONE function of this feature in the elbow joint.

_____

(1 mark)

(iv) People who suffer from arthritis experience pain because certain parts of the joint are worn away. Identify TWO parts of the joint that are likely to be worn away.

_____

_____

(2 marks)

(b) Figure 4 illustrates two mammalian vertebrae.

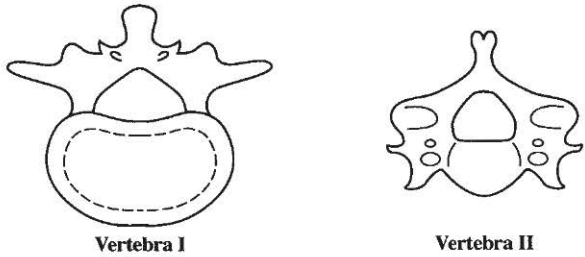

Figure 4. Diagram of two mammalian vertebrae

(i) Name EACH vertebra shown in Figure 4.

Vertebra I:

_____

Vertebra II:

_____

(2 marks)

(ii) Name THREE features that have similar functions in the two vertebrae shown in Figure 4.

_____

_____

_____

(3 marks)

(iii) Identify ONE feature that distinguishes between the two vertebrae.

_____

(1 mark)

Total 14 marks

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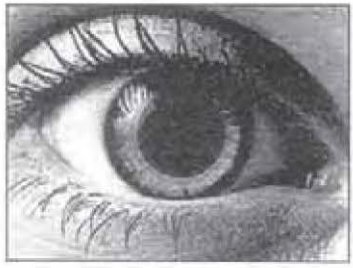
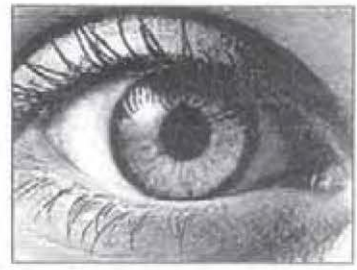
3. (a) The pupils of the human eye illustrated in Figure 5 show different responses to external stimuli.



Pupil I



Pupil II

Figure 5. Pupils of the human eye responding to stimuli

(i) Identify the stimulus for EACH pupil in Figure 5.

Stimulus for Pupil I:

_____

Stimulus for Pupil II:

_____

(2 marks)

(ii) For Pupil II in Figure 5, describe how the response came about.

_____

_____

_____

(3 marks)

(iii) How are the responses illustrated by the pupils in Figure 5 important to humans?

_____

_____

(2 marks)

(b) The diagram of the eye at Figure 6 shows how the lens appears in a *short-sighted* person.

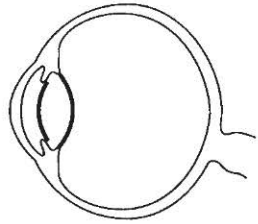
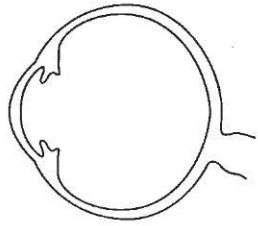


Figure 6. Diagram of the eye of a short-sighted person

(i) Using arrows, show on Figure 6 where an image that is viewed by this person would normally fall in relation to the retina. **(2 marks)**

(ii) Innovations in treating eye defects include lens replacement. On the diagram below, draw the lens that is likely to correct the defect shown in Figure 6, if lens replacement is done. **(2 marks)**



(c) Although plants have no specialized organs of sensitivity such as the eye, they also respond to external stimuli. Explain TWO ways in which the response in plants is different from the response to external stimuli by humans.

_____

_____

_____

_____

(4 marks)

Total 15 marks

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4. (a) Figure 7 below shows the longitudinal section of a tomato fruit.

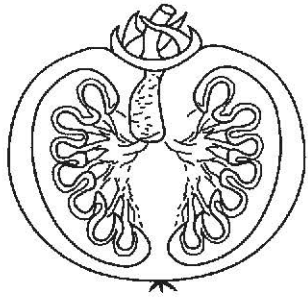


Figure 7. Longitudinal section of a tomato fruit

(i) Complete Figure 7 by labelling SIX parts of the fruit. **(6 marks)**

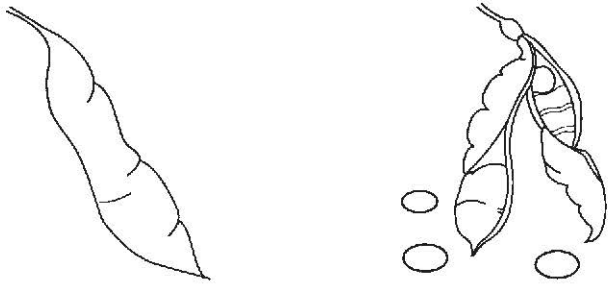
(ii) The seeds of the fruit shown in Figure 7 are dispersed by animals. Identify TWO ways in which these seeds are adapted for this process.

_____

_____

(2 marks)

(b) Figure 8 below shows the green and dry stages of a legume.



Green legume **Dry legume**

Figure 8. Green and dry stages of the same fruit of a legume

(i) What type of dispersal is shown by the fruit illustrated in Figure 8?

_____

(1 mark)

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(ii) Give TWO reasons why the plant parts illustrated in Figures 7 and 8 are classified by biologists as fruits.

_____

_____

_____

_____

(4 marks)

(c) Suggest ONE similarity and TWO differences in the ways in which the ovary in humans and in plants functions.

Similarity:

_____

(1 mark)

Differences:

_____

_____

(2 marks)

Total 16 marks

5. (a) Complete Table 3 below by writing an example and the cause of EACH type of disease listed.

TABLE 3: TYPES OF DISEASES

Type of disease	Example	Cause
Pathogenic		
Deficiency		
Hereditary		
Physiological		

(8 marks)

(b) Figure 9 below gives some data on factors affecting the transmission of HIV and AIDS in the Caribbean.

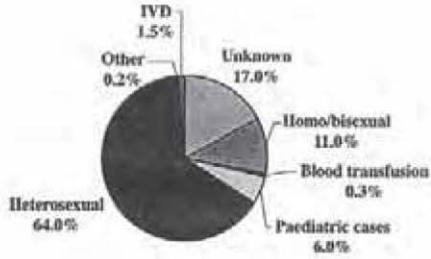

Figure 9. Factors affecting the transmission of HIV and AIDS in the Caribbean

(i) Using information from the pie chart, give the THREE most common ways in which HIV and AIDS are spread in the Caribbean.

_____

_____

_____

(3 marks)

(ii) Suggest TWO ways in which people in the Caribbean can reduce the spread of HIV and AIDS.

_____

_____

(2 marks)

(iii) Identify TWO symptoms of AIDS.

_____

_____

(2 marks)

Total 15 marks

END OF TEST